Characterization of 405 nm UV-Curable SLA Resin as a Photoresist for Silicon Photolithography

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Research question: How do spin-coating parameters control the thickness, wet-film stability, and adhesion of a 405 nm SLA resin on silicon?

Background

The Sun Devil Semiconductor Club is building a student-run photolithography process with 405 nm exposure. Consumer SLA resin is a candidate resist because it is:

- **Low cost:** ~$37/kg list price, far below commercial photoresists
- **Simple to process:** cures at the club's 405 nm wavelength and develops in IPA
- **Tough when cured:** ABS-like and non-brittle, so it may suit harsher processes or thicker etch masks

A grey resin (Siraya Tech Fast) is used because clear resin exposed poorly on our setup. It is spun uncured onto acetone-cleaned Si coupons in a non-cleanroom enclosure.

Objectives

- Measure how long the wet film resists dewetting after spinning, which sets how long a coated wafer can wait before exposure
- Establish the final speed for a good cured film, which bounds the usable process window
- Measure cured thickness vs. spin speed, and the thickness lost in the IPA rinse, so a target resist thickness can be dialed in

Wet-film imaging setup

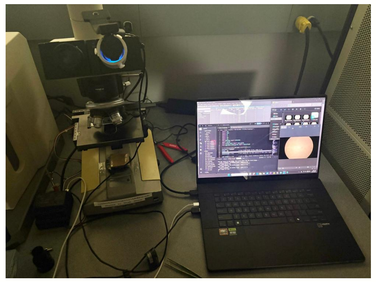


Fig. 1: Olympus BH microscope with an eyepiece webcam (Microsoft LifeCam). The incandescent lamp is relay-switched on only during captures to limit heating and exposure. 3 frames per time point; ImageJ counts new dewetting sites.

Cure and develop: which films survive

Good to 1550 rpm, peeling from 1600 rpm, fully removed above 2100 rpm (14 runs, 6 coupons).

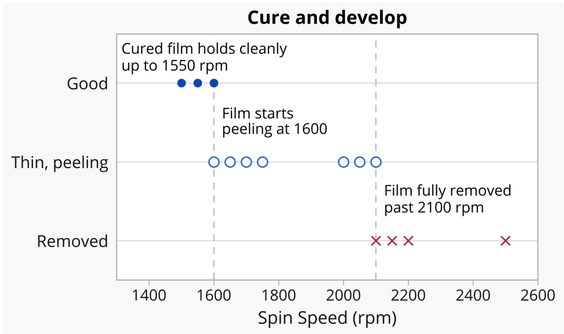


Fig. 2: 500 rpm spread 10 s, 10 s ramp, 60 s hold; then UV 20 s, IPA rinse ~10 s, UV 60 s.

Why thin films wash off

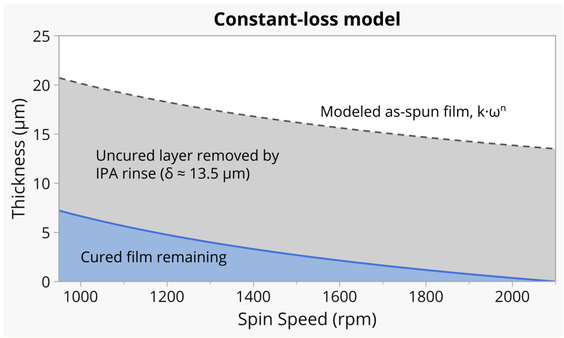


Fig. 3: Constant-loss model; the dashed as-spun curve is modeled, not measured.

The rinse removes an uncured top layer of fixed depth δ. Films thinner than δ wash off entirely, so no film remains above 2100 rpm.

Cured thickness vs. spin speed

6.1–7.3 µm at 1000 rpm down to 0.8 µm at 2000 rpm (6 coupons).

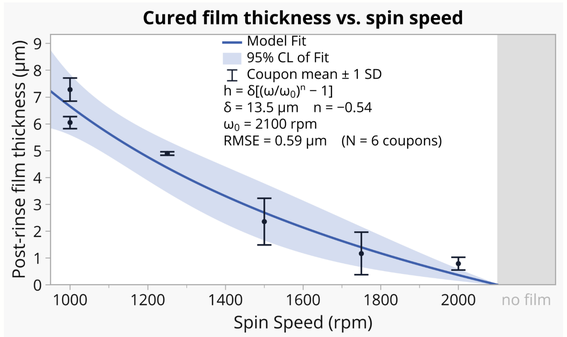


Fig. 4: Constant-loss fit with 95% confidence band (same process as Fig. 2).

$$h = \delta\left[\left(\frac{\omega}{\omega_0}\right)^{n} - 1\right]$$

As-spun film $\propto \omega^n$; the IPA rinse removes a fixed thickness δ. ω_0 = 2100 rpm is fixed from the cure-and-develop test (Fig. 2). δ = 13.5 µm, n = −0.54, RMSE 0.59 µm. δ and n are 0.9994 correlated, so the fit predicts thickness well but does not pin them down separately.

Stylus profilometry

Bruker Dektak XT, 3 mg stylus force. Kapton tape masks each coupon and is peeled after cure; scans start away from the resin ridge at the tape edge.

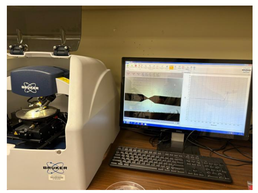
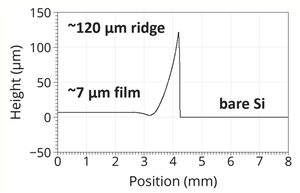


Fig. 5: Bruker Dektak XT. Fig. 6: Scan across the tape edge (1000 rpm): ~7 µm film plateau, ~120 µm edge ridge, bare Si.



Wet-film stability

Time to first dewetting site in the imaged field (club lab, 10 runs); "none" = none within 60 min.

Final speed	2 s ramp	10 s ramp	Runs
1500–2500 rpm	none	none	6
3500 rpm	0.5–1 min	15 min	4

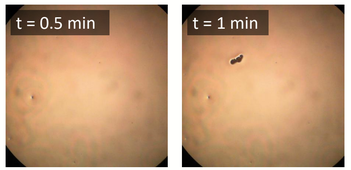
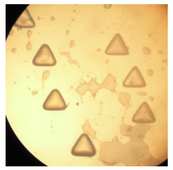


Fig. 7: Dewetting site forming on a wet film (3500 rpm, 2 s ramp).



Patterning

Fig. 8: Triangular features patterned in the resin on Si with the club's 405 nm setup. Shown for context; patterning was not optimized in this work.

Conclusions

- **≥ 60 min** wet-film stability at 1500–2500 rpm (5 of 6)
- **1550 rpm** max speed for a good cured film
- **~2.5–7 µm** cured thickness across 1000–1550 rpm
- **Fixed-depth** uncured layer lost in the IPA rinse

Limitations and next steps

Ellipsometry would not fit (~9 µm grey film absorbs and scatters); optical profilometry: film too thick. So n, k at 405 nm were not extracted.

One run per setting in a non-cleanroom lab, so dewetting results are relative to this environment; the imaged field covers only part of each coupon.

Next: replicates, larger-area inspection, and a cleanroom comparison.

Acknowledgments

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FURI

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